

Charity Nkemjika Umoren

Medical Doctor turned AI Engineer

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HEADLINE

Medical Doctor turned AI Engineer | LLMs, RAG & AI Agents | LangChain · LangGraph · CrewAI · n8n | Python · Full-Stack (React/Supabase) | Building reliable, useful AI systems

ABOUT

I'm a medical doctor turned AI engineer. I build LLM-powered applications, agentic systems, RAG pipelines, and end-to-end automations that connect AI models to real work.

I design and build across the full AI engineering stack — from multi-agent workflows and conversational AI to full-stack apps and business automations. Recent hands-on work includes RAG pipelines (LangChain + ChromaDB), multi-agent systems with LangGraph, CrewAI and AG2/AutoGen, a Gradio-based MCP server toolkit, a resume-tailoring app running on both OpenAI and Gemini, and n8n automations for reporting and operations. I run models locally (PyTorch, HuggingFace, Ollama/Qwen) on my own GPU workstation and ship production web apps with React, Supabase and Netlify.

At the Medical AI Laboratory (MAI Lab) in Lagos, I work at the intersection of engineering, research operations and communications — coordinating AI cancer-imaging projects, preprocessing and analysing medical images for machine learning, managing a research team, and turning complex work into clear reporting and outreach.

What sets me apart is that I don't just implement AI — I understand the operational and clinical context around it. Medicine trained me to work in high-stakes, detail-critical environments, and that shows up in how I build: reliable, well-structured systems that are actually useful.

Core stack: Python · LangChain · LangGraph · CrewAI · n8n · OpenAI & Gemini APIs · HuggingFace · PyTorch · Ollama · vector databases (ChromaDB) · RAG · scikit-learn · SQL · React · Supabase · Netlify · Streamlit · Power BI.

Open to remote AI Engineer, ML Engineer, LLM Engineer and Applied AI roles across any industry. Let's connect.

EXPERIENCE

Medical Artificial Intelligence Laboratory (MAI Lab) — Lab Manager & Director of Community Relations

Nov 2024 – Present · Lagos State, Nigeria

- Coordinate AI cancer screening and diagnostics projects, including machine learning for low-cost medical image analysis; preprocess and analyse breast ultrasound (POCUS) and MRI images for ML pipelines.
- Manage the African Breast Imaging Dataset (ABreast) — curating point-of-care ultrasound scans matched with mammograms and clinical data, with ~450 women screened onsite to date across sites in Nigeria and Uganda.
- Lead a team of 5 researchers and interns across 6 concurrent projects, owning timelines, data management and reporting.
- Coordinate cross-functional collaboration between clinicians, data scientists and engineers to translate care needs into working AI solutions.
- Run community breast-cancer screening outreaches across 18 communities in Nigeria and Uganda, screening 1,000+ women (Count Me In).

- Lead technical communications and content: LinkedIn campaigns, conference content (ISMIRM 2026, Cape Town), the AI in Medicine (AIM) Fellowship call and application system, and partner coordination.

Lagos University Teaching Hospital — House Officer

Nov 2023 – Oct 2024 · Lagos State, Nigeria

- Conducted patient assessments (histories, physical exams, diagnostic tests) for ~20 patients per week and collaborated with multidisciplinary teams on treatment plans.
- Maintained accurate patient records and delivered patient and family education on conditions and preventive care.
- Ran a data-analysis project on rheumatoid-arthritis patients seen in clinic.

VOLUNTEER EXPERIENCE

Hope365 Network — Operations & Full-Stack Developer (Volunteer)

Nigeria

- Designed and shipped Hope365 Workspace, a production project-management web app (React, Supabase, Netlify serverless functions) with a full invite-and-email system, role-based access, account self-deletion with admin safeguards, and branded transactional email templates.
- Handle operations, coordination and reporting for the NGO.

Hope Nation — Administrative & Data Coordinator (Volunteer, Remote)

Manchester, United Kingdom

- Built a monthly attendance-analytics pipeline from Zoom exports (raw CSV → attendance matrix → departmental breakdown → formatted report with charts).
- Produced post-event reporting (multi-volume decks with financial slides and native charts) and coordinated logistics for major ministry events.

PROJECTS

- **Hope365 Workspace** — Full-stack project-management app (React · Supabase · Netlify). Invite-and-email system, role-based access, admin-safeguarded account deletion.
- **AI Engineering Notebooks** — 13 documented Jupyter notebooks covering RAG, agents, multi-model systems and ML.
- **AI Automation Workflows** — n8n workflows for reporting and operational automation.
- **NeuroAfrica Dashboard** — Streamlit + Plotly dashboard over a 316-study African neuroimaging scoping review, with 8 analytical views (trends, geography, disease landscape, imaging tech, demographics, collaboration, research quality, tables).
- **Resume Tailoring App** — LLM application on OpenAI + Gemini with normalised structured outputs across both SDKs.
- **MCP Server Toolkit** — Gradio-based Model Context Protocol tools.

SKILLS

LLMs · Retrieval-Augmented Generation (RAG) · AI Agents · LangChain · LangGraph · CrewAI · AG2/AutoGen · n8n · Model Context Protocol (MCP) · OpenAI API · Gemini API · HuggingFace · Ollama · Prompt Engineering · Vector Databases · ChromaDB · Gradio · Streamlit · Python · PyTorch · scikit-learn · TensorFlow · Keras · pandas · Machine Learning · Predictive Modelling · Feature Engineering · Model Evaluation · Medical Image Analysis (MRI/Ultrasound) · Data Analysis · Data Visualisation · SQL · Power BI · React · Supabase · Netlify · Full-Stack Development · Git · Serverless Functions · API Integration · Technical Communication · Research Operations · Team Leadership · Project Management

EDUCATION

- **University of Uyo** — Bachelor of Medicine, Bachelor of Surgery (MBBS), 2014 – 2023
- **ExploreAI Academy** — Data Science Certification, 2023 – 2024
- **ALX Africa** — ALX Foundations (Foundations in Tech) & Data Science, 2023 – 2024